

CENTA Carbon Fibre Shafts

Type S for High Speed and/or Long Spans / Baureihe S für hohe Drehzahlen und/oder lange Spannweiten

HIGH-SPEED & LONG SPAN OPTIMIZED SHAFT SYSTEMS

CENTA carbon fibre composite shafts Type S are specifically engineered to accommodate ultra-high rotational speed boundaries and exceptionally long distance spans without intermediate support requirements. Utilizing precision glass and carbon laminate compositions, Type S optimizes structural deflection behaviors for sensitive architectural installations such as cooling towers and marine drivelines.

KEY ADVANTAGES

- **Optimized for Long Spans:** Engineered structure enables maximum distance-between-shaft designs while strictly avoiding critical speed resonance.
- **Laminate with High Speed Suitability:** Fiber placement angles focus specifically on high dynamic stability and minimal radial expansion under centrifugal forces.
- **Severe Weight Reduction:** Up to 70% lighter than standard heavy steel configurations, easing operational strain on surrounding coupling infrastructure.
- **Corrosion and Environmental Shielding:** Robust material composite construction limits rust or structural wear under extreme humidity and temperature drops.

TECHNICAL DRAWING & CONSTRUCTION LAYOUT

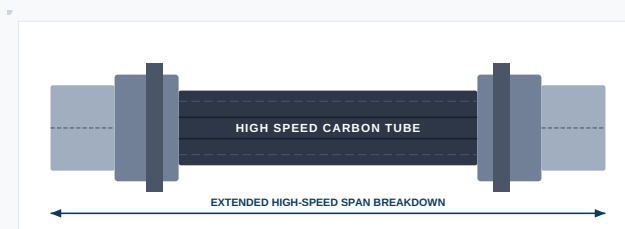


Figure 1: Sectional layout drawing of CENTA Type S high-speed/long span composite shaft assembly, detailed with metallic flange endpoints and specialized light-mass center tube core.

TECHNICAL SPECIFICATIONS TABLE

Complete catalog performance ratings and dimensions for Baureihe S (Series S).

Shaft Size	Nominal Torque [kNm]	O.D. [mm]	Mass per meter length [kg]	Mass moment of inertia / m length [kgm ²]	Torsional stiffness / m length [MNm/rad]	Mass per end fitting* [kg]	Intermediate shaft bearing Ø [mm]	Nominal Ø of bulkhead seal [mm]
Laminate with high speed suitability								
CFRP5S167	5	167	5,7	0,036	0,046	5,6	75	200
CFRP6S169	6	169	6,5	0,042	0,053	5,6	80	200
CFRP8S216	8	216	6,9	0,076	0,095	7,2	85	260
CFRP10S219	10	219	8,5	0,094	0,118	7,2	95	260
CFRP12S267	12	267	8,0	0,136	0,170	12	100	290
CFRP15S270	15	270	9,9	0,171	0,213	12	105	320
CFRP20S316	20	316	11,7	0,277	0,347	18	120	350
CFRP25S319	25	319	13,9	0,334	0,418	22	125	350
CFRP30S323	30	323	17,0	0,413	0,516	22	135	380
CFRP40S372	40	372	18,8	0,614	0,768	45	150	110
CFRP50S378	50	378	24,2	0,802	1,002	45	160	410
CFRP60S384	60	384	29,6	0,999	1,248	45	170	410
CFRP80S435	80	435	34,7	1,513	1,890	41	190	470
CFRP100S442	100	442	41,9	1,863	2,328	48	200	470
CFRP120S533	120	533	40,4	2,699	3,372	98	210	560
CFRP150S540	150	540	49,3	3,340	4,173	98	230	560
CFRP200S615	200	615	57,8	5,114	6,390	150	250	650

* Mass per end fitting values relate to standard manufacturing configurations. Continuous updates and customizable alterations may adapt these indices over time.

Procurement & Technical Support in India:

For standard requests, precise engineering support, torque alignment calculation, or official quotations, please get in touch with our authorized engineering partner network:

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